

Antiwork Topic Modeling

Exploratory longitudinal analysis for collaborator review

97,254

filtered posts

199

non-outlier clusters

38.6%topic assignment
n=37,536 of 97,254**19.0%**theme-map coverage
n=18,506 of 97,254

Study summary

Across 577,190 archived r/antiwork submission records retrieved from March 2021 through February 2025, the management-term filter retained 97,254 unique posts for analysis. The updated BERTopic analysis identifies a descriptive shift away from COVID-centered health and safety discussion and toward scheduling and time-off discussion. Compensation, career precarity, and recruitment remain present across the four rolling windows.

Completed analysis

Rebuilt the analysis with BERTopic; exported document-level assignments and four rolling-window fits; finalized a 27-cluster, nine-theme codebook; and produced figures and reproducibility materials.

Interpretive scope

The figures show the share of all filtered posts captured by included candidate-theme clusters. The 27-cluster map is not a full taxonomy or a complete estimate of management-related content.

Selected findings

Theme	Y1 (n=28,276)	Y4 (n=11,960)	Interpretation
Health safety and attendance	4.97% (n=1,406)	2.50% (n=299)	Early COVID/illness concentration contracts.
Scheduling hours and time off	1.71% (n=484)	3.20% (n=383)	Largest mapped theme in Y4.
Compensation and pay rights	3.52% (n=994)	3.04% (n=363)	Remains substantial in every window after a Y2 high.
Career precarity and exit	2.54% (n=718)	2.47% (n=295)	Broadly stable across four rolling windows.

All percentages divide posts in included candidate-theme clusters by every filtered post in the same rolling window. They show the portion of the filtered corpus captured by this 27-cluster map, not a complete measure of all management-related content.

Data collection

Public r/antiwork submissions were retrieved through the Arctic Shift archive API in 48 monthly UTC batches from 1 March 2021 through 1 March 2025. This collection yielded 577,190 archived submission records before filtering. The analysis retained 97,254 unique posts whose title or body matched the established management terms boss, manager, supervisor, or team lead, including plural variants. The resulting corpus is an archive-based sample of public discourse, not a representative sample of employees or an author-level panel. Archive records may differ from content displayed on Reddit after collection.

Analytic workflow

The analysis proceeded as follows. Corpus checks, model-based topic discovery, descriptive prevalence estimates, and robustness evidence are reported separately.

- 1. Collect, filter, and audit the corpus.** Retrieved 577,190 archived submission records across 48 monthly pulls, then applied the management-term filter and confirmed 97,254 unique retained post IDs. A duplicate audit found 1.2% residual duplicate non-placeholder texts; this is reported rather than silently removed.
- 2. Construct the primary text field.** Used post title plus body after handling deleted and placeholder content. A title-only run was retained as a sensitivity analysis, not substituted for the primary representation.
- 3. Discover topics with a current topic-modeling pipeline.** Embedded documents with all-MiniLM-L6-v2, reduced the embedding space with UMAP, clustered with HDBSCAN, and represented topics with BERTopic c-TF-IDF. The full-corpus model produced 199 non-outlier clusters and assigned 38.6% of posts.
- 4. Document the theme map.** Reviewed the largest 30 clusters and retained 27 interpretable clusters in a nine-theme candidate codebook. Generic, deleted, and community-meta clusters were excluded, yielding 18,506 mapped posts (19.0% of the full filtered corpus).
- 5. Estimate descriptive longitudinal patterns.** Calculated each theme's share of every filtered post in four March-to-February rolling windows. The figures in this brief show the main monthly patterns, the full reviewed-theme comparison, and monthly assignment coverage.
- 6. Test stability and state the boundary.** Ran seed-stability checks, title-only sensitivity, and one-to-one top-term alignment checks across rolling-window fits. A fixed RoBERTa whole-post polarity check then scored all 97,254 retained posts and was joined to the reviewed map without refitting either model; causal, human-validated sentiment, and full-conversation prevalence claims remain out of scope.

The private repository includes the analysis scripts, codebook, diagnostics, and aggregate evidence used in this brief. Raw posts and text-level samples remain local.

From collection to estimates

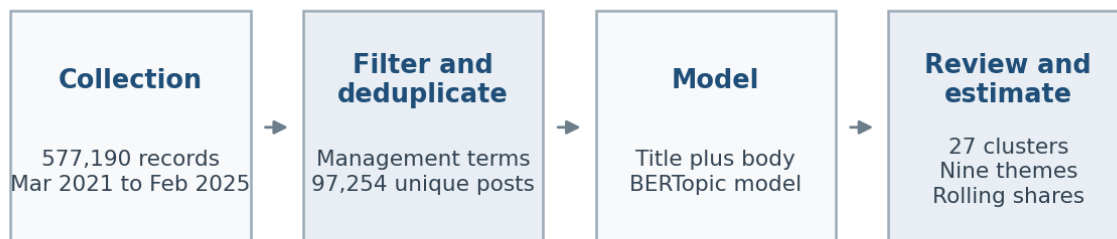
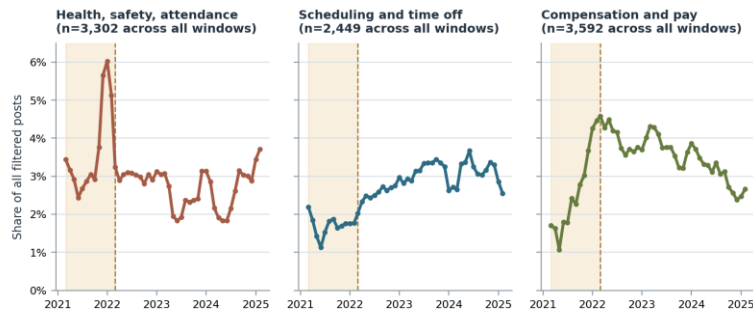


Figure 1. The collection, filtering, modeling, review, and estimation sequence used for the exploratory analysis.

Monthly patterns in the main themes



Three-month rolling shares; shading marks the Y1 Great Resignation window (Mar 2021-Feb 2022) and the dashed line marks March 2022. Denominator is all filtered posts each month; panel n values are theme counts across all windows.

Figure 2. Three-month rolling prevalence of the themes that anchor the main interpretation. Shares use all filtered posts each month as the denominator.

Month-by-month longitudinal patterns in the ten largest reviewed clusters

The top-level candidate map contains nine themes, so a literal top-ten theme chart would be mislabeled. For the requested month-by-month longitudinal view, the figure below shows the ten largest included BERTopic clusters, nested within the nine-theme map and labeled as subthemes rather than as additional themes. Each panel uses the full filtered monthly corpus as its denominator.

Month-by-month prevalence of the ten largest reviewed subthemes

Exact monthly shares of all filtered posts. Shading marks the Y1 Great Resignation window (Mar 2021-Feb 2022); dashed line marks March 2022.

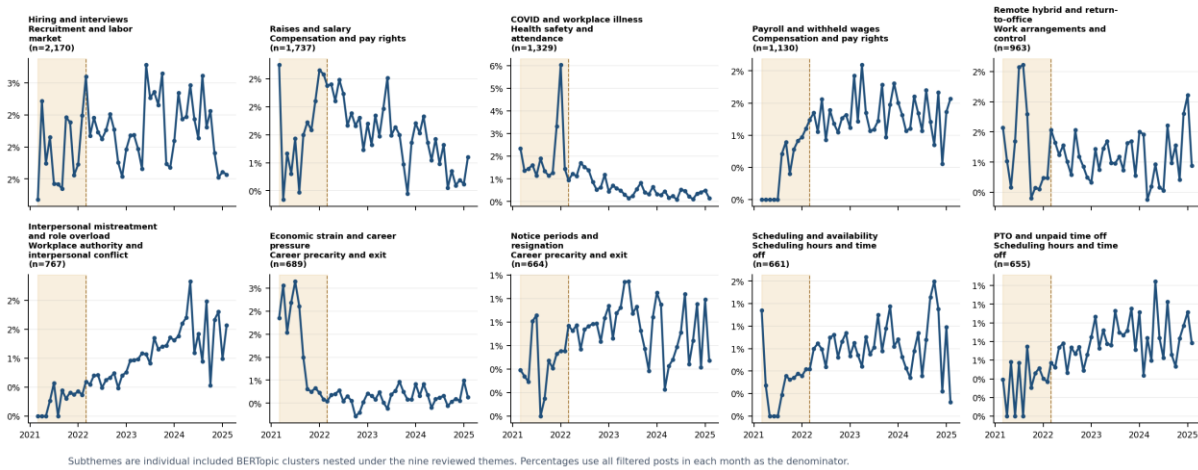


Figure 3. Month-by-month longitudinal prevalence of the ten largest reviewed clusters, with each cluster's parent candidate theme in its label. The shaded period is the Y1 Great Resignation window, March 2021 through February 2022. Panel *n* values are full-period cluster sizes.

All reviewed themes by rolling window

Provisional theme prevalence by rolling window

Share of all filtered posts; rolling-window denominators are shown in the legend. Exploratory 27-cluster map excludes generic, deleted, and community-meta clusters.

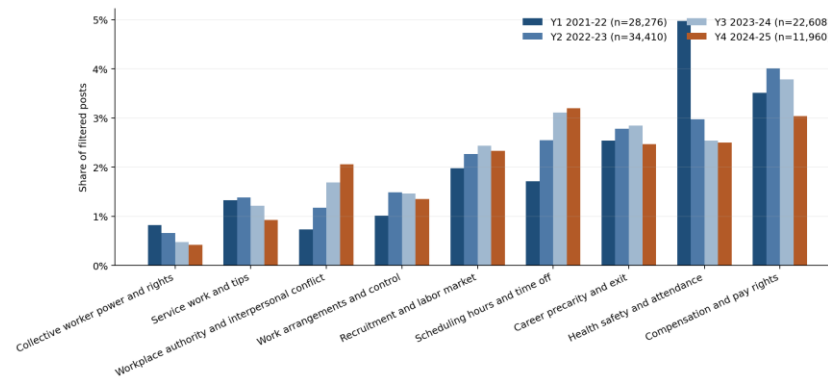


Figure 4. Exploratory prevalence by rolling window. The legend gives each window denominator; Table 1 gives the numerator n behind every displayed percentage. The 27-cluster map excludes generic, deleted, and community-meta clusters.

Counts behind rolling-window percentages

Each cell gives the theme share of all filtered posts in that rolling window followed by its numerator n. The column-header n is the all-filtered-post denominator for that window.

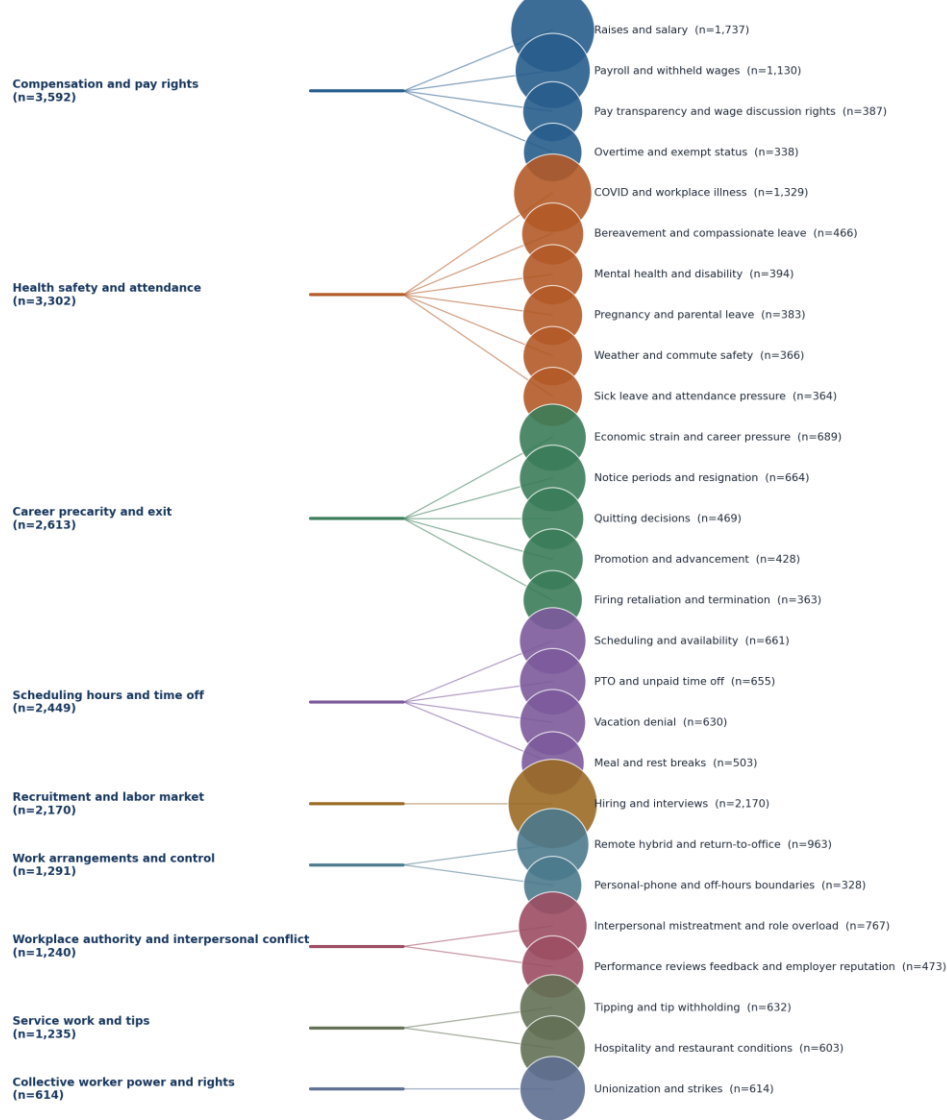
Candidate theme	Y1 2021-22 (n=28,276)	Y2 2022-23 (n=34,410)	Y3 2023-24 (n=22,608)	Y4 2024-25 (n=11,960)
Compensation and pay rights	3.52% (n=994)	4.01% (n=1,379)	3.79% (n=856)	3.04% (n=363)
Health safety and attendance	4.97% (n=1,406)	2.97% (n=1,023)	2.54% (n=574)	2.50% (n=299)
Career precarity and exit	2.54% (n=718)	2.78% (n=956)	2.85% (n=644)	2.47% (n=295)
Scheduling hours and time off	1.71% (n=484)	2.55% (n=878)	3.11% (n=704)	3.20% (n=383)
Recruitment and labor market	1.98% (n=560)	2.27% (n=780)	2.44% (n=551)	2.33% (n=279)
Work arrangements and control	1.01% (n=286)	1.49% (n=513)	1.46% (n=330)	1.35% (n=162)
Workplace authority and interpersonal conflict	0.74% (n=208)	1.17% (n=404)	1.69% (n=382)	2.06% (n=246)
Service work and tips	1.32% (n=374)	1.38% (n=476)	1.21% (n=274)	0.93% (n=111)
Collective worker power and rights	0.82% (n=231)	0.66% (n=226)	0.47% (n=107)	0.42% (n=50)

Table 1. Counts and prevalence for every reviewed candidate theme across the four rolling windows. Values describe the documented 27-cluster map, not all management-related discussion.

Reviewed theme hierarchy

Reviewed candidate-theme hierarchy

Theme totals and subtheme cluster sizes are calculated from full-corpus BERTopic document assignments.



Included clusters only: 27 clusters, 18,506 posts (19.0% of the filtered corpus). Node area scales with cluster n. This is a documented candidate map, not a complete taxonomy.

Figure 5. The nine candidate themes, their included subthemes, and full-corpus cluster sizes. Node area scales with n; this documented map covers 18,506 posts, or 19.0% of the filtered corpus.

Exploratory Y1 and Y4 comparison

The comparison below contrasts the user-defined Great Resignation period, March 2021 through February 2022, with the most recent March 2024 through February 2025 window. It is a descriptive comparison of the reviewed theme map, not an estimate of the Great Resignation's effect.

Candidate-theme prevalence in Y1 and Y4

Exploratory comparison of the user-defined Great Resignation period (Mar 2021-Feb 2022) with Mar 2024-Feb 2025. Lines show percentage-point change, not causal effects.

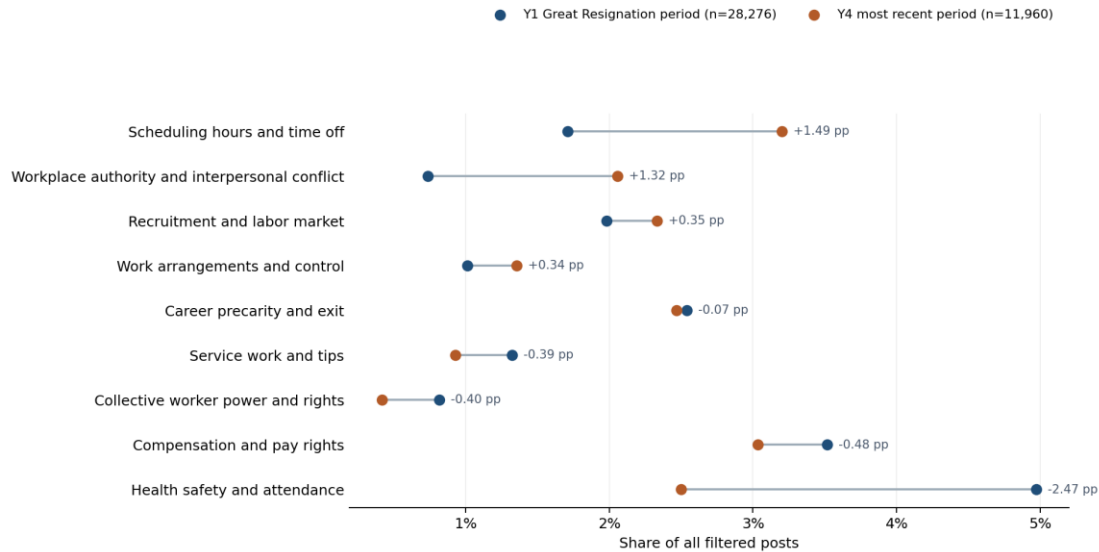


Figure 6. Candidate-theme prevalence in Y1 and Y4. The legend gives the all-filtered-post denominator for each window; Table 1 gives each theme numerator n ; labels show descriptive percentage-point differences.

Whole-corpus polarity check

Context-aware model

A fixed social-media RoBERTa classifier estimated whole-post negative, neutral, and positive polarity across every retained post. This is a contextual transformer-based estimate, rather than a lexicon method such as VADER.

Interpretive boundary

The check is model-based, not human-validated or targeted specifically to management. It tests whether the intentionally problem-oriented sample is predominantly negative under this classifier; it does not estimate average employee sentiment.

Across 97,254 scored posts, the model's hard labels were 62.9% negative, 30.4% neutral, and 6.7% positive. Mean posterior probabilities were 54.1%, 35.7%, and 10.2%, respectively. Mean maximum class probability was 70.0%; 16.4% of posts reached the 512-token limit.

The hard negative-label share was 59.8% (n=16,923 of 28,276) in Y1, the Great Resignation window, and 67.7% (n=8,096 of 11,960) in Y4. This descriptive increase is not a causal estimate and does not establish a change in employee sentiment.

Whole-corpus RoBERTa sentiment by rolling window

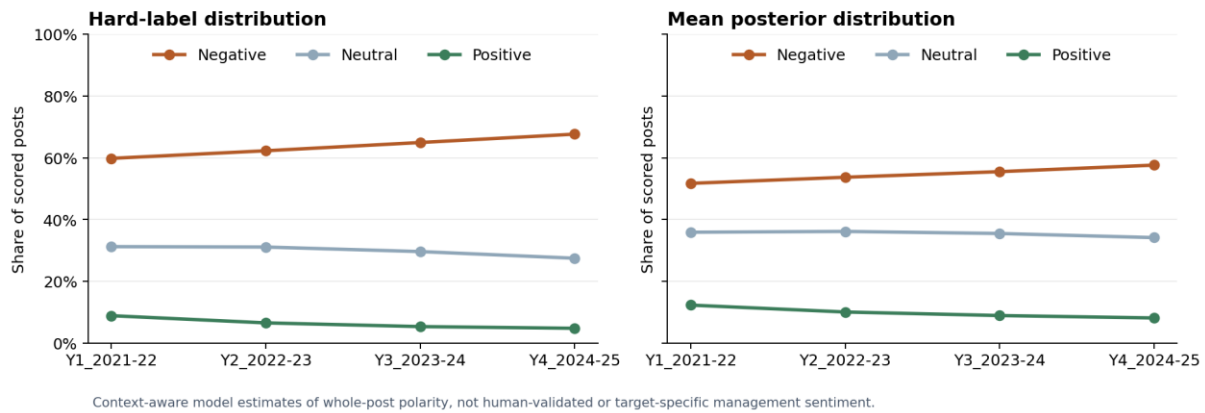


Figure 7. Whole-corpus RoBERTa polarity estimates across the four rolling windows. The hard-label and mean-posterior views are shown together because model uncertainty is material to interpretation.

Polarity within reviewed themes

The figure below joins the fixed post-level RoBERTa estimates to the existing reviewed BERTopic map. It does not refit either model. Values describe whole-post negative-polarity probability among posts assigned to each theme, while the prevalence figures above retain all filtered posts as their denominator.

Modeled negative-polarity probability within reviewed themes

Three-month rolling mean of post-level RoBERTa negative probability. Shading marks the Y1 Great Resignation window (Mar 2021-Feb 2022); dashed line marks March 2022. The thematic map covers 19.0% of the filtered corpus.

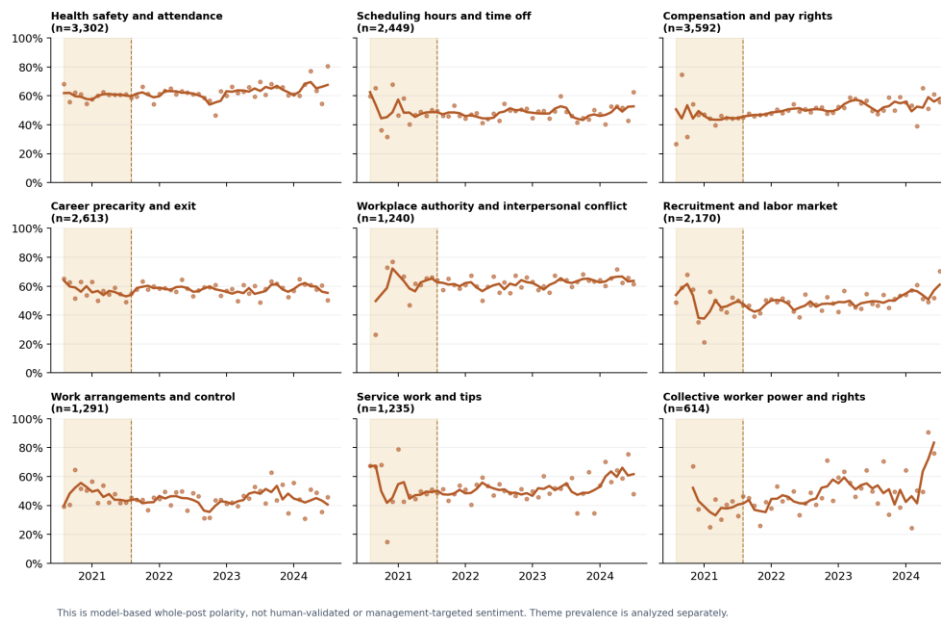


Figure 8. Three-month rolling mean of modeled negative-polarity probability within the nine reviewed themes. The theme map covers 18,506 posts, or 19.0% of the filtered corpus.

Analytic approach and scope

Primary analysis

Sentence-transformer embeddings (all-MiniLM-L6-v2), UMAP, HDBSCAN, BERTopic topic representations, and a full-corpus model for longitudinal prevalence. Per-window fits serve as stability diagnostics.

Coverage of estimates

The model assigns 38.6% of posts to non-outlier clusters. The final theme map covers 18,506 posts: 19.0% of all filtered posts and 49.3% of assigned posts.

Assignment coverage

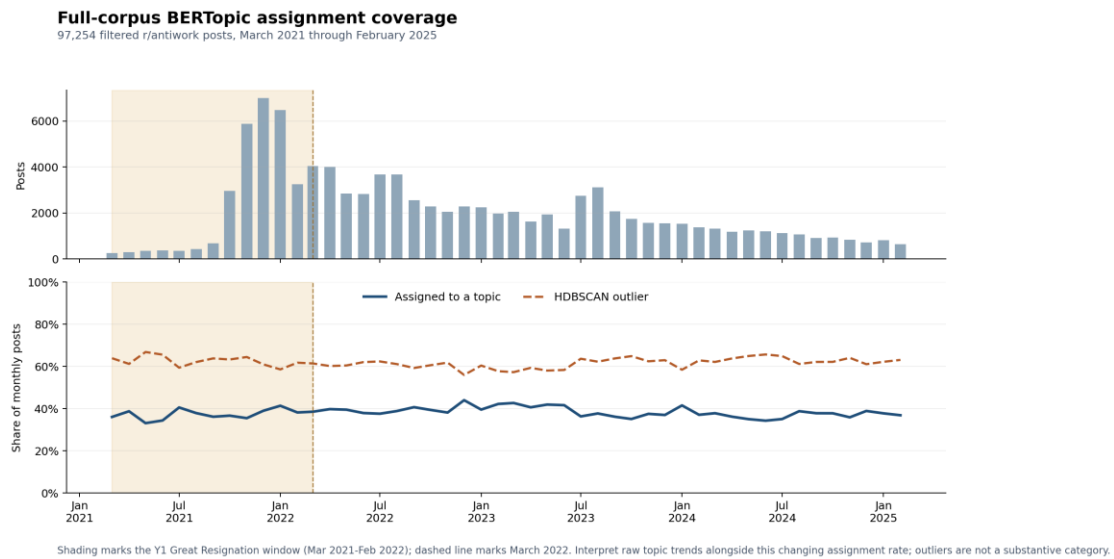


Figure 9. Assignment coverage varied from 33.1% to 44.0% by month. Outliers are a coverage limitation, not a substantive category.

Robustness checks

Check	Result
Corpus audit	Inputs contain 97,254 matching unique post IDs; the management-term filter re-matches 100% of retained posts.
Seed stability	Across three UMAP seeds on the same 10,000-post stratified sample, joint-inlier ARI was 0.900 to 0.966.
Window alignment	Conservative one-to-one term matching is supporting context, not proof of substantive persistence.
Text sensitivity	Titles-only clustering was stable but materially different from title-plus-body; title-plus-body remains the primary representation.

A larger all-mpnet-base-v2 encoder check was attempted but did not complete within the available CPU bound, so it is deferred and not treated as evidence.

Interpreting the results

Observed patterns

Within the mapped themes, health, safety, and attendance discussion declined over the four windows, while scheduling, hours, and time-off discussion increased. Compensation, career precarity, and recruitment remained visible throughout.

What the results support

The topic patterns describe changes in the composition of management-related public discussion and identify domains for follow-up with representative employee data. They do not establish employee sentiment, turnover drivers, or policy effects.

Remaining work

Available for drafting

Results, figures, a theme codebook, coverage diagnostics, seed stability, text-mode sensitivity, whole-corpus polarity estimates, and a defined interpretive scope are ready for the manuscript.

Future validation

The planned human-validation study below would test construct validity beyond the current exploratory evidence. It has not been conducted and is not a basis for any current result.

Planned hybrid human-validation study

The strongest next step is a two-stage hybrid validation study, not a single preset-topic yes-or-no check. The design separates discovery of human-meaningful concepts from direct evaluation of the frozen BERTopic theme map, while keeping coders blinded to model assignments during primary coding.

1. Blinded inductive coding. Draw a stratified exploratory subsample across the major reviewed themes, HDBSCAN outliers or general posts, and the four rolling years. Two trained coders independently identify the dominant concern in each post without being shown the model topic or candidate theme. Approximately 30 to 50 posts per major theme, plus outlier and general-post strata, is a practical starting range; the final sample size should be set prospectively based on feasible coding capacity.
2. Freeze a human-readable codebook. Use the first sample to document missing concepts, merged concepts, inclusion and exclusion rules, and examples. Freeze the nine-theme codebook before drawing a fresh, independent validation sample. Validate the nine top-level themes before attempting the more granular 27-cluster subtheme map.
3. Independently validate the model map. On the fresh sample, coders apply the frozen codebook while still blinded to BERTopic assignments. Report raw agreement and Krippendorff's alpha before adjudication. Then compare adjudicated human codes with the BERTopic map using per-theme precision, recall, F1, confusion matrices, and coverage of outlier or other posts.
4. Use preset fit ratings only as a secondary audit. After independent coding is complete, representative posts and random nonmatching controls may be rated for how well a displayed model label fits. This can assess coherence and interpretability, but it should not replace blind coding because showing the proposed theme first can anchor raters.

Manuscript revision status

Sections revised

The revised manuscript includes the title and abstract; current-study framing and research question; Methods, Results, Discussion, implications, limitations; figures; primary BERTopic references; and the collection-to-filtering count of 577,190 retrieved archive records followed by 97,254 analyzed posts.

Before circulation

Add the completed polarity method and its bounded exploratory results to the manuscript, then harmonize retained Background language with the paper's descriptive scope. Historical theory and prior turnover findings should remain context, not claims tested by this study.

Sampling-frame framing

Intended analytic focus

The management-term filter intentionally concentrates a problem-oriented, likely negatively skewed subset of r/antiwork discussion. The purpose was to surface recurring management concerns that may warrant follow-up.

Boundary for the manuscript

Do not present the results as average employee sentiment or a standalone priority ranking. Frame the themes as candidate areas to examine alongside representative evidence and organizational context.

Materials for drafting

Draft in this order: collection and sampling frame; coverage, theme-prevalence, and model-based polarity results; limitations; then an exploratory discussion. Use the documented theme map and avoid causal, full-conversation, or human-validated sentiment claims.

Reproducible code and shareable evidence: <https://github.com/tristowwww/antiwork-topic-modeling>

Method records: planning/SCRAPING.md, planning/PREPROCESSING.md, planning/WRITING_HANDOFF.md, planning/ROBUSTNESS.md, and planning/CANDIDATE_THEME_CODEBOOK.csv.

Current status: the topic-modeling and full-corpus polarity analyses are complete. The manuscript is revised as a descriptive, exploratory topic-prevalence study and needs the completed polarity method and results incorporated before circulation.

Appendix: Reviewed cluster map

The table below makes the review trail visible. It lists all 30 largest clusters examined for the candidate-theme map, their full-corpus size (n), representative terms, assigned candidate theme and subtheme, and inclusion decision.

Reviewed clusters 1-15

ID	n	Representative terms	Candidate theme and subtheme	Decision
0	2,170	interview hiring interviews recruiter applied resume	Recruitment and labor market Hiring and interviews	Include high confidence
1	2,078	bad good hate conversation like worst	Other Generic/deleted manager posts	Exclude high confidence
2	1,737	raise raises increase salary inflation pay	Compensation and pay rights Raises and salary	Include high confidence
3	1,477	rehearsal good team bad employees people	Other Generic manager-role discussion	Exclude medium confidence
4	1,329	covid positive tested test sick symptoms	Health safety and attendance COVID and workplace illness	Include high confidence
5	1,130	check paycheck paid pay deposit payroll	Compensation and pay rights Payroll and withheld wages	Include high confidence
6	963	remote office home hybrid wfh remotely	Work arrangements and control Remote hybrid and return-to-office	Include high confidence
7	767	coworker like things said just	Workplace authority and interpersonal conflict	Include

		doesn	Interpersonal mistreatment and role overload	medium confidence
8	689	life school want career college money	Career precarity and exit Economic strain and career pressure	Include medium confidence
9	664	notice weeks period week giving resignation	Career precarity and exit Notice periods and resignation	Include high confidence
10	661	schedule availability scheduled saturday sunday days	Scheduling hours and time off Scheduling and availability	Include high confidence
11	655	pto vacation use approved days request	Scheduling hours and time off PTO and unpaid time off	Include high confidence
12	632	tips tip tipping restaurant pool cash	Service work and tips Tipping and tip withholding	Include medium confidence
13	630	vacation request trip days approved time	Scheduling hours and time off Vacation denial	Include high confidence
14	614	union unions strike unionize workers unionizing	Collective worker power and rights Unionization and strikes	Include high confidence

Reviewed clusters 16-30

ID	n	Representative terms	Candidate theme and subtheme	Decision
15	603	restaurant kitchen chef cook food bar	Service work and tips Hospitality and restaurant conditions	Include high confidence
16	503	break breaks lunch minute 30 minutes	Scheduling hours and time off Meal and rest breaks	Include high confidence
17	473	review reviews performance glassdoor feedback rating	Workplace authority and interpersonal conflict Performance reviews feedback and employer reputation	Include medium confidence
18	469	quit quitting resign leaving tell today	Career precarity and exit Quitting decisions	Include high confidence
19	466	funeral bereavement died mom passed family	Health safety and attendance Bereavement and compassionate leave	Include high confidence
20	428	promotion promoted position role team level	Career precarity and exit Promotion and advancement	Include high confidence
21	394	mental health autistic disability autism disabled	Health safety and attendance Mental health and disability	Include high confidence
22	387	wages discuss discussing illegal nlrp pay	Compensation and pay rights Pay transparency and wage discussion rights	Include high confidence
23	383	pregnant pregnancy baby maternity leave paternity	Health safety and attendance Pregnancy and parental leave	Include high confidence
24	366	snow weather storm roads hurricane ice	Health safety and attendance Weather and commute safety	Include high confidence
25	364	sick calling called come days feeling	Health safety and attendance Sick leave and attendance pressure	Include high confidence
26	363	fired firing got adverse threatened activity	Career precarity and exit Firing retaliation and termination	Include medium confidence
27	338	overtime exempt ot hours salaried 40	Compensation and pay rights Overtime and exempt status	Include high confidence
28	328	phone phones personal cell app number	Work arrangements and control Personal-phone and off-hours boundaries	Include high confidence
29	321	antiwork sub anti movement subreddit society	Other Community meta-discussion	Exclude high confidence